

AJAY GARG

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B.Tech, Computer Science and Engineering - Cloud Computing
CGPA: 8.58/10 | Delhi, India

Summary

Final-year B.Tech CSE (Cloud Computing) student at SRMIST with a CGPA of 8.58, building production-grade systems across full-stack web development, AI/ML, agentic AI, and DevOps. Experienced in shipping end-to-end products — from multi-service LLM orchestration pipelines and Docker-based microservices to JWT-secured REST APIs and CI/CD-deployed containerised applications. Actively seeking Software Development internship opportunities.

Education

SRM IST-Ghaziabad	2027
B.Tech · Computer Science and Engineering - Cloud Computing	CGPA - 8.58/10
Arwachin International School	2022
Class XII - CBSE · Delhi	Percentage - 81.5%
Arwachin International School	2020
Class X - CBSE · Delhi	Percentage - 85.5%

Projects

VEDA - Agentic AI Personal Operating System	Jan 2025 – Present
Personal · Python, FastAPI, Docker, Groq/Llama 3.3 70B, FAISS, GitHub Actions, Nginx	Agentic AI
Architected a microservices-based agentic OS with four FastAPI services (Orchestrator, Planner, Memory, Executor) orchestrated via Docker Compose and Nginx. Built an LLM planning engine integrating Groq-hosted Llama 3.3 70B with a provider abstraction layer that converts natural language into dependency-aware task graphs using Kahn's topological sort. Implemented a two-layer memory system (FAISS + fastembed embeddings over SQLite) for semantic long-term context retrieval. Authored GitHub Actions CI/CD pipelines pushing multi-service Docker images to Docker Hub on every push.	
Email AI Agent	AI
Personal · FastAPI, LangChain, Gmail API (OAuth 2.0), LLMs	
Built an LLM-powered agent using LangChain that autonomously summarises, categorises, and generates contextual replies for incoming emails. Implemented full OAuth 2.0 token flow with Gmail API for secure delegated mailbox access. FastAPI backend with persistent storage exposes RESTful endpoints for cross-device access, with retry logic handling LLM and API rate-limit failures gracefully.	
Digital Vault - Secure Personal Data Management Platform	2024
Personal · FastAPI, React/TypeScript, PostgreSQL, JWT, Agora SDK, Railway, Vercel	Full-Stack
Developed a full-stack secure data management platform with JWT access token + refresh token rotation (industry-standard stateless auth pattern). Integrated Agora SDK for real-time peer-to-peer video calling. Deployed FastAPI backend to Railway and React/TypeScript frontend to Vercel. Features include file storage, project management, notes, and live video — all with deliberate, documented architecture trade-offs.	
Celestia	Web App
Freelance · FastAPI, MySQL, React, OpenAI GPT, Razorpay, SMTP	
Built a full-stack astrology SaaS using FastAPI, MySQL, and React with OTP-based login alongside JWT session management. Designed a deterministic rule-based astrology computation engine (Swiss Ephemeris) enhanced with GPT-generated AI explanations, cleanly separating deterministic logic from generative output. Integrated Razorpay with subscription and one-time appointment flows including webhook-driven payment state management.	
Crisp	Backend
Personal · Node.js, Express, PostgreSQL, React	

Designed and implemented a self-hosted Firebase-inspired Backend-as-a-Service system — providing reusable authentication primitives and real-time data APIs consumable by multiple front-end clients. Architected PostgreSQL schema with connection-pooling strategy to support concurrent application workloads without per-app infrastructure overhead.

Skills

Programming Languages: Python, Java, JavaScript, Go

Backend: FastAPI, Node.js/Express, Flask · REST API Design, JWT/OAuth 2.0, WebSockets, Microservices

Frontend: React.js, TypeScript, Tailwind CSS, HTML/CSS

AI / ML: LangChain, LLM Integration (Groq, OpenAI), FAISS, Vector Embeddings, Agentic Pipelines

Databases: PostgreSQL, MySQL, MongoDB, SQLite, FAISS (vector store)

DevOps & Cloud: Docker, Docker Compose, Nginx, GitHub Actions (CI/CD), Railway, Vercel, Linux/Bash

Tools: Git, Postman, VS Code, JWT Authentication

Achievements & Extras

Co-authored academic research paper on autonomous self-healing cloud infrastructure — Prometheus/Grafana monitoring, Autoencoder-based anomaly detection, and Docker-based automated remediation workflows.

Frontend contributor, AI Compliance Management System — Code Wizards 2.0 Hackathon, SRMIST; built React UI with dark Mission Control aesthetic integrated with Qdrant vector DB via FastAPI backend.

Studying advanced DSA in Java: topological sort (Kahn's), DSU with path compression, interval DP, sliding window, Floyd's cycle detection — targeting competitive SDE compensation benchmarks.

Languages

English [Professional Working Proficiency], Hindi [Professional Working Proficiency]

Links

[LinkedIn](#), [GitHub](#), [LeetCode](#)